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The equation $x^4 + x + 6y^3 - y - 1 = 0$ has no integer solutions

Size $h = 69$, length $l = 11.58$

Theorem 1. *The equation*

$$x^4 + x + 6y^3 - y - 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses a polynomial identity of the form $HT = U^2 + 3V^2$ on the curve (1), and the fact that an odd prime p with $(-3/p) = -1$ divides $U^2 + 3V^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm form $a^2 + 3b^2$

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right)\left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(-3/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + 3b^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv -3(bc)^2 \equiv -3 \pmod{p}$, so -3 is a square modulo p . As $(-3/p) \neq 0$, it is a non-zero square, so $(-3/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 + 3b^2 \neq 0$, and $v_p(a^2 + 3b^2)$ is even for every odd prime p with $(-3/p) = -1$.*

Proof. Clearly $a^2 + 3b^2 > 0$. Let p be an odd prime with $(-3/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + 3b'^2$ by Lemma 2. Since $a^2 + 3b^2 = p^{2k}(a'^2 + 3b'^2)$, we get $v_p(a^2 + 3b^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{12}$. Then $(-3/n') = (-3/n)$.*

Proof. Write $-3 = \varepsilon 2^e q$ with $\varepsilon = -1$, $e = 0$ and $q = 3$. By multiplicativity and the reciprocity laws,

$$\left(\frac{-3}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since 12 is divisible by $4q$, the claim follows. $\square$

Lemma 5. *Let n be a non-zero integer, and let s , u and m be positive integers such that every prime factor of s divides 6, both $2s$ and $12s$ divide m , u is odd, $(-3/u) = -1$, and $n \equiv su \pmod{m}$. Suppose also that $n > 0$. Then there is an odd prime p with $(-3/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by 12, n' is odd and $n' \equiv u \pmod{12}$. Since $n > 0$, Lemma 4 gives $(-3/n') = (-3/u) = -1$. By multiplicativity, the product of $(-3/p)^{v_p(n')}$ over the prime factors p of n' is -1 , so some prime factor p of n' satisfies $(-3/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 6$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + x + 6y^3 - y - 1$, so that (1) is the equation $F(x, y) = 0$. Define

$$H = 4x^2 + 12xy + 2x + 36y^2 + 18y + 4,$$

$$\begin{aligned} T = & 3359232x^5 - 10077696x^4y + 5785344x^4 - 2239488x^3y + 746496x^3 \\ & - 4478976x^2y + 6158592x^2 + 16796160xy^2 - 13996800xy + 186624x \\ & - 1679616y^2 + 6718464y - 2892672, \end{aligned}$$

$$U = 7776x^4 + 1728x^3 - 10368x^2 - 3888x + 4320,$$

$$V = -7776x^3 - 4752x^2 + 5616x - 1296,$$

$$\begin{aligned} G = & -60466176x^4 - 13436928x^3 + 6718464x^2 + 100776960xy - 36951552x \\ & - 10077696y + 35271936, \end{aligned}$$

$$A = 1535310x^2 + 1737517x - 97905,$$

$$B = 1535310x^3 + 1140452x^2 - 1346978x - 1715173.$$

Expanding both sides, one checks the identities

$$HT - U^2 - 3V^2 = GF, \tag{2}$$

$$AU + BV = 2^4 \cdot 3^5 \cdot 457 \cdot 1013, \tag{3}$$

$$12H = 3(2x + 12y + 3)^2 + (6x - 1)^2 + 20. \tag{4}$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) gives

$$HT = U^2 + 3V^2, \tag{5}$$

where H , T , U and V now denote the values of these polynomials at (x, y) .

Step 1: congruences. Checking the 2592^2 residue pairs modulo 2592, we find that (1) has exactly 2592 solutions modulo 2592, and that at each of them H is congruent modulo 2592 to one of the 186 residues r listed in Table 1. Each r is listed there as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(-3/u) = -1$. Here every prime factor of each $s \in \{2, 8, 18, 54, 72, 216\}$ divides 6, and $2s$ and $12s$ divide 2592 for each such s .

Step 2: signs. By (4), $H > 0$.

Step 3: primes p with $(-3/p) = -1$. Let p be an odd prime with $(-3/p) = -1$. We show that $v_p(H)$ is even. By (3), U and V are not both zero, so (5) and Lemma 3 show that $v_p(HT) = v_p(U^2 + 3V^2)$ is even. Suppose that p divides both H and T . Then $p \mid U^2 + 3V^2$ by (5), so $p \mid U$ and $p \mid V$ by Lemma 2, and $p \mid 2^4 \cdot 3^5 \cdot 457 \cdot 1013$ by (3). As p is odd and $p \neq 3$ and $p \neq 457$ because $(-3/3) = 0$ and $(-3/457) = 1$, we get $p = 1013$. Checking the 1013^2 residue pairs modulo 1013, we find that F , H and T have no common zero modulo 1013, a contradiction. Hence p does not divide both H and T . If $p \nmid T$, then $v_p(H) = v_p(HT)$ is even, and if $p \nmid H$, then $v_p(H) = 0$.

Step 4: contradiction. By Step 1, $H \equiv su \pmod{2592}$ with s and u as in Lemma 5, and $H > 0$ by Step 2. Hence Lemma 5 applies to $n = H$, so there is an odd prime p with $(-3/p) = -1$ such that $v_p(H)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The residues $r = su$ of H modulo 2592 in Step 1.

$10 = 2 \cdot 5$	$466 = 2 \cdot 233$	$898 = 2 \cdot 449$	$1306 = 2 \cdot 653$	$1762 = 2 \cdot 881$	$2194 = 2 \cdot 1097$
$34 = 2 \cdot 17$	$472 = 8 \cdot 59$	$904 = 8 \cdot 113$	$1330 = 2 \cdot 665$	$1768 = 8 \cdot 221$	$2200 = 8 \cdot 275$
$40 = 8 \cdot 5$	$490 = 2 \cdot 245$	$922 = 2 \cdot 461$	$1336 = 8 \cdot 167$	$1786 = 2 \cdot 893$	$2218 = 2 \cdot 1109$
$58 = 2 \cdot 29$	$514 = 2 \cdot 257$	$946 = 2 \cdot 473$	$1354 = 2 \cdot 677$	$1810 = 2 \cdot 905$	$2242 = 2 \cdot 1121$
$82 = 2 \cdot 41$	$520 = 8 \cdot 65$	$952 = 8 \cdot 119$	$1378 = 2 \cdot 689$	$1816 = 8 \cdot 227$	$2248 = 8 \cdot 281$
$88 = 8 \cdot 11$	$522 = 18 \cdot 29$	$954 = 18 \cdot 53$	$1384 = 8 \cdot 173$	$1818 = 18 \cdot 101$	$2250 = 18 \cdot 125$
$90 = 18 \cdot 5$	$538 = 2 \cdot 269$	$970 = 2 \cdot 485$	$1386 = 18 \cdot 77$	$1834 = 2 \cdot 917$	$2266 = 2 \cdot 1133$
$106 = 2 \cdot 53$	$562 = 2 \cdot 281$	$994 = 2 \cdot 497$	$1402 = 2 \cdot 701$	$1858 = 2 \cdot 929$	$2290 = 2 \cdot 1145$
$130 = 2 \cdot 65$	$568 = 8 \cdot 71$	$1000 = 8 \cdot 125$	$1426 = 2 \cdot 713$	$1864 = 8 \cdot 233$	$2296 = 8 \cdot 287$
$136 = 8 \cdot 17$	$586 = 2 \cdot 293$	$1018 = 2 \cdot 509$	$1432 = 8 \cdot 179$	$1882 = 2 \cdot 941$	$2314 = 2 \cdot 1157$
$154 = 2 \cdot 77$	$594 = 54 \cdot 11$	$1042 = 2 \cdot 521$	$1450 = 2 \cdot 725$	$1890 = 54 \cdot 35$	$2338 = 2 \cdot 1169$
$178 = 2 \cdot 89$	$610 = 2 \cdot 305$	$1048 = 8 \cdot 131$	$1474 = 2 \cdot 737$	$1906 = 2 \cdot 953$	$2344 = 8 \cdot 293$
$184 = 8 \cdot 23$	$616 = 8 \cdot 77$	$1066 = 2 \cdot 533$	$1480 = 8 \cdot 185$	$1912 = 8 \cdot 239$	$2362 = 2 \cdot 1181$
$202 = 2 \cdot 101$	$634 = 2 \cdot 317$	$1080 = 216 \cdot 5$	$1498 = 2 \cdot 749$	$1930 = 2 \cdot 965$	$2376 = 216 \cdot 11$
$226 = 2 \cdot 113$	$658 = 2 \cdot 329$	$1090 = 2 \cdot 545$	$1522 = 2 \cdot 761$	$1954 = 2 \cdot 977$	$2386 = 2 \cdot 1193$
$232 = 8 \cdot 29$	$664 = 8 \cdot 83$	$1096 = 8 \cdot 137$	$1528 = 8 \cdot 191$	$1960 = 8 \cdot 245$	$2392 = 8 \cdot 299$
$250 = 2 \cdot 125$	$682 = 2 \cdot 341$	$1114 = 2 \cdot 557$	$1546 = 2 \cdot 773$	$1978 = 2 \cdot 989$	$2410 = 2 \cdot 1205$
$274 = 2 \cdot 137$	$706 = 2 \cdot 353$	$1138 = 2 \cdot 569$	$1570 = 2 \cdot 785$	$2002 = 2 \cdot 1001$	$2434 = 2 \cdot 1217$
$280 = 8 \cdot 35$	$712 = 8 \cdot 89$	$1144 = 8 \cdot 143$	$1576 = 8 \cdot 197$	$2008 = 8 \cdot 251$	$2440 = 8 \cdot 305$
$298 = 2 \cdot 149$	$730 = 2 \cdot 365$	$1162 = 2 \cdot 581$	$1594 = 2 \cdot 797$	$2026 = 2 \cdot 1013$	$2458 = 2 \cdot 1229$
$306 = 18 \cdot 17$	$738 = 18 \cdot 41$	$1170 = 18 \cdot 65$	$1602 = 18 \cdot 89$	$2034 = 18 \cdot 113$	$2466 = 18 \cdot 137$
$322 = 2 \cdot 161$	$754 = 2 \cdot 377$	$1186 = 2 \cdot 593$	$1618 = 2 \cdot 809$	$2050 = 2 \cdot 1025$	$2482 = 2 \cdot 1241$
$328 = 8 \cdot 41$	$760 = 8 \cdot 95$	$1192 = 8 \cdot 149$	$1624 = 8 \cdot 203$	$2056 = 8 \cdot 257$	$2488 = 8 \cdot 311$
$346 = 2 \cdot 173$	$778 = 2 \cdot 389$	$1210 = 2 \cdot 605$	$1642 = 2 \cdot 821$	$2074 = 2 \cdot 1037$	$2506 = 2 \cdot 1253$
$360 = 72 \cdot 5$	$792 = 72 \cdot 11$	$1224 = 72 \cdot 17$	$1656 = 72 \cdot 23$	$2088 = 72 \cdot 29$	$2520 = 72 \cdot 35$
$370 = 2 \cdot 185$	$802 = 2 \cdot 401$	$1234 = 2 \cdot 617$	$1666 = 2 \cdot 833$	$2098 = 2 \cdot 1049$	$2530 = 2 \cdot 1265$
$376 = 8 \cdot 47$	$808 = 8 \cdot 101$	$1240 = 8 \cdot 155$	$1672 = 8 \cdot 209$	$2104 = 8 \cdot 263$	$2536 = 8 \cdot 317$
$394 = 2 \cdot 197$	$826 = 2 \cdot 413$	$1242 = 54 \cdot 23$	$1690 = 2 \cdot 845$	$2122 = 2 \cdot 1061$	$2538 = 54 \cdot 47$
$418 = 2 \cdot 209$	$850 = 2 \cdot 425$	$1258 = 2 \cdot 629$	$1714 = 2 \cdot 857$	$2146 = 2 \cdot 1073$	$2554 = 2 \cdot 1277$
$424 = 8 \cdot 53$	$856 = 8 \cdot 107$	$1282 = 2 \cdot 641$	$1720 = 8 \cdot 215$	$2152 = 8 \cdot 269$	$2578 = 2 \cdot 1289$
$442 = 2 \cdot 221$	$874 = 2 \cdot 437$	$1288 = 8 \cdot 161$	$1738 = 2 \cdot 869$	$2170 = 2 \cdot 1085$	$2584 = 8 \cdot 323$